

Invertible Matrices, Adjoins, Row Reduction, and Rotations

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§1.1 Invertibility

A square matrix A is invertible if there exists a matrix A^{-1} such that

$$A^{-1}A = I, \quad AA^{-1} = I.$$

Some immediate consequences recorded in the note are:

- (i) If A is invertible, then Gaussian elimination has a pivot in every column and every row.
- (ii) A may have many distinct powers, but it has only one inverse.
- (iii) If A is invertible, then the equation $Ax = b$ has the unique solution

$$x = A^{-1}b.$$

- (iv) If the homogeneous equation $Ax = 0$ has a nonzero solution, then A is not invertible.

The last point is fundamental: invertibility is equivalent to trivial kernel for a square matrix.

§1.2 The 2×2 formula

For a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

we have

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

provided

$$ad - bc \neq 0.$$

Thus the determinant is the obstruction to invertibility.

§1.3 Diagonal matrices

If

$$A = \text{diag}(d_1, \dots, d_n),$$

then A is invertible iff every $d_i \neq 0$, and

$$A^{-1} = \text{diag}(d_1^{-1}, \dots, d_n^{-1}).$$

This is the simplest case where the inverse is completely visible.

§1.4 Products of inverses

For invertible matrices,

$$(AB)^{-1} = B^{-1}A^{-1}.$$

More generally,

$$(ABC)^{-1} = C^{-1}B^{-1}A^{-1}.$$

The order reverses because composition reverses when undoing operations.

§1.5 Adjugate formula

Let $A = (a_{ij})$. Delete row i and column j to get the minor M_{ij} . The cofactor is

$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$

The adjugate matrix is the transpose of the cofactor matrix:

$$\text{adj}(A) = (C_{ji}).$$

Then

$$A^{-1} = \frac{1}{\det A} \text{adj}(A)$$

when $\det A \neq 0$.

The formula is computationally expensive but conceptually important: it proves that entries of the inverse are rational functions of entries of A .

§1.6 Row reduction for inverses

The practical method is row reduction:

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

For example, for

$$M = \begin{pmatrix} 1 & 4 \\ 2 & 9 \end{pmatrix},$$

one has

$$\det M = 9 - 8 = 1,$$

and hence

$$M^{-1} = \begin{pmatrix} 9 & -4 \\ -2 & 1 \end{pmatrix}.$$

The note also illustrates obtaining this inverse by elementary row operations on the augmented matrix.

§1.7 Rotation matrices

The final inserted page discusses planar rotations. The rotation matrix is

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

It sends

$$\begin{pmatrix} x \\ y \end{pmatrix}$$

to

$$\begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{pmatrix}.$$

Its inverse is

$$R_\theta^{-1} = R_{-\theta} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

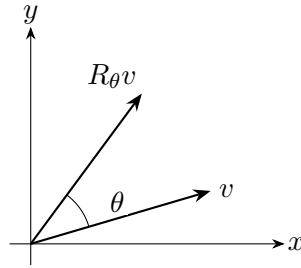


Figure 1.1: A planar rotation acts linearly on vectors and is represented by R_θ .

§1.8 Beyond the first calculation

Invertible matrices are automorphisms of finite-dimensional vector spaces. Gaussian elimination, determinants, adjugates, and block decompositions are different ways to detect or compute the same object: a reversible linear transformation.

§1.9 Invertibility as isomorphism

A matrix is invertible exactly when the associated linear map is an isomorphism. This means three equivalent things in finite dimensions:

$$\ker A = 0, \quad \operatorname{im} A = V, \quad \det A \neq 0.$$

Row reduction tests the same fact by constructing inverse operations.

§1.10 Adjugate and exterior powers

The adjugate formula

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$$

comes from cofactors. Conceptually, determinants describe the action of A on top exterior power, while cofactors encode how $(n-1)$ -dimensional volume elements transform.

§1.11 Rotations as orthogonal matrices

A rotation matrix R satisfies

$$R^T R = I, \quad \det R = 1.$$

The first condition preserves lengths and angles; the second preserves orientation. Thus rotations are the orientation-preserving part of the orthogonal group.

§1.12 Invertibility, volume, and orthogonal motion

Inverse formulas, row reduction, adjugates, diagonal matrices, and rotations are all different faces of isomorphism: solve uniquely, preserve or rescale volume, and understand what the map does independently of coordinates.

§1.13 Inverse as undoing a linear process

The inverse matrix should be read as an operation that undoes a linear process. The formula

$$(AB)^{-1} = B^{-1}A^{-1}$$

then becomes obvious: if one first applies B and then A , one must undo A first and then undo B . Row reduction, adjugates, and determinants are three different languages for this same reversibility.

The rotation matrix is a particularly good example because the inverse is visible geometrically. Rotating by θ is undone by rotating by $-\theta$, so

$$R_\theta^{-1} = R_{-\theta} = R_\theta^T.$$

This is the first glimpse of orthogonal matrices, where algebraic inverse equals geometric transpose.

§1.14 Invertible matrices as a Lie group

The set of all invertible $n \times n$ matrices over $\mathbb{R}$ is

$$GL(n, \mathbb{R}) = \{A : \det A \neq 0\}.$$

It is a group under multiplication and also an open subset of $\mathbb{R}^{n^2}$. Therefore it is a Lie group: a group with a smooth manifold structure.

The tangent space at the identity is the Lie algebra

$$\mathfrak{gl}(n, \mathbb{R}) = M_n(\mathbb{R}).$$

The exponential map

$$\exp X = I + X + \frac{X^2}{2!} + \cdots$$

connects infinitesimal generators to invertible transformations. Elementary matrix powers and exponentials are thus the entry point to continuous symmetry.

§1.15 Determinant as a volume form

The determinant is not merely a formula for invertibility. It is the factor by which a linear map scales oriented volume:

$$\text{Vol}(Av_1, \dots, Av_n) = \det(A) \text{Vol}(v_1, \dots, v_n).$$

The multiplicative identity

$$\det(AB) = \det A \det B$$

says that volume scaling composes multiplicatively.

This is why row operations affect determinants in exactly the familiar ways. Swapping rows reverses orientation, scaling a row scales volume, and adding one row to another shears volume without changing it.

§1.16 Polar decomposition

Every invertible real matrix has a polar decomposition

$$A = QP,$$

where Q is orthogonal and P is symmetric positive definite. One can take

$$P = (A^T A)^{1/2}, \quad Q = AP^{-1}.$$

Geometrically, P stretches along orthogonal principal axes, and Q rotates or reflects. This separates deformation into “pure strain” and “rigid motion.”

The elementary inverse/rotation examples become clearer through this decomposition. Orthogonal matrices preserve lengths; positive definite matrices perform anisotropic stretching; a general invertible matrix is both.

§1.17 Numerical stability of inverses

Although A^{-1} exists when $\det A \neq 0$, computing it may be unstable. The condition number

$$\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

measures how much errors may be amplified. A tiny determinant is not by itself the best diagnostic; singular values describe directional collapse.

Thus the elementary criterion $\det A \neq 0$ answers existence, while numerical linear algebra asks sensitivity.

§1.18 Row operations and the general linear group

Gaussian elimination writes an invertible matrix as a product of elementary matrices. Geometrically, elementary matrices are simple transformations: shears, scalings, and swaps. Thus an invertible linear transformation can be built from elementary geometric moves. This is the computational skeleton behind the group $GL(n)$.

The determinant map

$$\det : GL(n, \mathbb{R}) \rightarrow \mathbb{R}^\times$$

is a group homomorphism. Its kernel is the special linear group

$$SL(n, \mathbb{R}) = \{A : \det A = 1\}.$$

This group consists of volume-preserving linear transformations. Elementary determinant tests therefore separate invertible maps by how they scale volume.

For rotations, the relevant subgroup is

$$O(n) = \{Q : Q^T Q = I\}, \quad SO(n) = O(n) \cap SL(n, \mathbb{R}).$$

Orthogonal matrices preserve the inner product, hence all lengths and angles. The chain

$$SO(n) \subset O(n) \subset GL(n)$$

organizes the original examples: rotations are the most rigid, orthogonal maps allow reflections, and invertible maps allow arbitrary linear distortion without collapse.

§1.19 Exponential map and infinitesimal motion

A smooth path $A(t)$ in $GL(n)$ with $A(0) = I$ has derivative $X = A'(0)$, an arbitrary matrix. Conversely, a matrix X generates the path

$$A(t) = e^{tX}.$$

If X is skew-symmetric, then e^{tX} is orthogonal. Indeed,

$$\frac{d}{dt}(e^{tX})^T e^{tX} = 0$$

when $X^T = -X$. This gives a precise meaning to “skew-symmetric matrices generate rotations.”

The elementary rotation matrices in the note are therefore not isolated formulas; they are one-parameter subgroups of $GL(n)$.